

Mammalia Raccoon Proximity Network Analysis

<https://github.com/aiyshwary/Raccoon-Network-Analysis>

Python

NetworkX

Jupyter

Graph Analysis

Epidemiology

OVERVIEW

A graph-based epidemiological study modelling rabies transmission risk in urban raccoon populations through proximity network analysis and centrality measures, supporting public health and wildlife management decisions.

IMPACT

Constructed and analyzed a raccoon proximity network using real-world contact data, applying centrality measures to identify high-risk individuals and quantify potential rabies spread pathways within urban populations.

KEY DETAILS

- Built a weighted proximity network from the mammalia-raccoon-proximity dataset using Python and NetworkX.

- Computed degree, betweenness, closeness, and eigenvector centrality to rank individuals by transmission risk.

- Identified superspreader nodes whose removal would most disrupt disease propagation chains.

- Visualized network topology and centrality distributions to communicate findings clearly.

- Results provide evidence-based recommendations for targeted wildlife intervention strategies.